EAST WEST UNIVERSITY

CSE347:System Analysis and Design
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Group No: 03
(Section No. 01)

Project Report

Smart E-Complaint & Public Service Management System

Submitted by:

Student ID	Student Name
Al Zarin	2023-3-60-454
Joy Dutta	2023-3-60-463
Sumya Kawsar	2023-3-60-168
Riyadh Hossain	2023-3-60-262
Sanzida Islam Shoha	2023-3-60-009

Submitted to:

Dr. Hosney Jahan
Lecturer
Department of CSE
East West University

Introduction

In many countries, including developing regions, public service complaint handling is still largely dependent on traditional manual processes. These systems are often inefficient, time-consuming and lack transparency, resulting in delayed responses and dissatisfaction among citizens. Paper-based complaint submissions make it difficult to track progress, maintain records and ensure accountability of responsible authorities.

With the rapid growth of digital technology and e-governance initiatives, there is a strong need to modernize public service systems. Citizens now expect faster communication, real-time updates and easy access to government services through online platforms. However, many existing systems still fail to provide a unified, user-friendly solution for submitting and monitoring complaints.

To address these challenges, the Smart E-Complaint & Public Service Management System has been developed. This system provides an integrated digital platform where citizens can submit complaints related to public services, track their status in real time, and provide feedback after resolution. At the same time, government officers can efficiently manage, process, and resolve complaints, while administrators oversee the entire system.

The project aims to improve transparency, enhance efficiency, and reduce the limitations of traditional complaint handling methods by leveraging modern web-based technologies.

Literature Review

The development of the Smart E-Complaint & Public Service Management System is inspired by various existing studies and systems in the field of e-governance and digital public service delivery.

Many governments around the world have adopted e-governance systems to improve communication between citizens and public authorities. These systems aim to enhance transparency, accountability and efficiency in handling public service requests and complaints. Research shows that digital platforms significantly reduce processing time and improve citizen satisfaction compared to traditional methods.

Traditional complaint handling systems in government offices are mostly manual and paper-based, which often leads to several issues such as loss of records, delayed responses, lack of proper tracking and limited accountability. Studies highlight that these limitations reduce public trust in government services and create inefficiencies in administrative processes.

Several countries have introduced web-based and mobile complaint management applications where citizens can submit issues related to utilities, transportation, sanitation and other public

services. These systems provide features like complaint registration, status tracking and feedback mechanisms. However, many of these platforms lack proper integration, priority handling or user-friendly design, which affects their overall effectiveness.

Recent research also emphasizes the importance of real-time tracking systems and priority-based complaint management**,** which help authorities identify critical issues quickly and allocate resources more effectively. Additionally, incorporating a citizen feedback system has been found to improve service quality and accountability.

Based on these existing systems and research findings, this project proposes a more integrated and efficient solution that combines complaint submission, tracking, prioritization and feedback into a single platform, aiming to overcome the limitations of previous approaches.

Problem Statement

In many public service sectors, the process of handling citizen complaints is still inefficient, slow and lacks transparency. Most government offices rely on manual or semi-digital systems, where complaints are submitted through paper forms or unstructured channels. This leads to several critical issues, including loss of information, delays in processing, poor communication between citizens and authorities and lack of accountability.

Citizens often face difficulties in tracking the progress of their complaints, which creates frustration and reduces trust in public service institutions. At the same time, government officials struggle to manage a large number of complaints effectively due to the absence of a centralized and organized system. Urgent issues are not always prioritized properly and feedback from citizens is rarely collected or utilized to improve service quality.

Furthermore, existing digital solutions in some regions are either limited in functionality, not user-friendly or lack integration between different departments. This results in fragmented systems that fail to deliver an efficient and reliable complaint management process.

Therefore, there is a need for a smart, centralized and user-friendly system that allows citizens to submit complaints easily, track their status in real time and enables authorities to manage, prioritize and resolve issues efficiently while ensuring transparency and accountability.

Objectives of the Project

The main objective of the Smart E-Complaint & Public Service Management System is to develop an efficient and transparent digital platform for handling public service complaints.

Specific Objectives:

- To provide an easy and user-friendly system for citizens to submit complaints online.
- To enable real-time tracking of complaint status for better transparency.
- To reduce manual, paper-based processes and improve overall efficiency.
- To help government officers manage, process and resolve complaints effectively.
- To implement a priority-based system for handling urgent and critical issues.
- To improve communication between citizens and public authorities.
- To collect feedback from citizens to enhance service quality.
- To maintain a centralized database for better record-keeping and monitoring.

Methodology

The Smart E-Complaint & Public Service Management System is designed as a web-based platform that streamlines the process of complaint submission, management and resolution through a centralized and automated system.

The proposed system follows a structured methodology consisting of the following components:

1. System Architecture

The system is based on a client-server architecture, where users interact through a web interface (frontend) and all data processing and storage are handled by the backend server and database. This ensures scalability, security and efficient data management.

2. User Roles and Access Control

The system defines three main user roles:

- **Citizens:** Submit complaints, track status and provide feedback.
- **Government Officers:** View, manage and resolve assigned complaints.
- **Administrator:** Manage users, departments and overall system operations.

Each user has role-based access to ensure security and proper system control.

3. Complaint Submission Module

Citizens can submit complaints online by providing necessary details such as category, description, location and priority level. The system stores this data in a centralized database.

4. Complaint Processing and Assignment

Once a complaint is submitted:

- It is automatically categorized and assigned to the relevant department or officer.
- A unique complaint ID is generated for tracking purposes.
- Priority levels (low, medium, high) help in handling urgent issues efficiently.

5. Real-Time Tracking System

Users can track the status of their complaints (e.g., submitted, in progress, resolved) in real time, ensuring transparency and reducing uncertainty.

6. Feedback and Evaluation Module

After resolution, citizens can provide feedback, which helps evaluate service quality and improve system performance.

7. Database Management

All complaints, user data and system records are stored in a centralized database, ensuring proper record-keeping, easy retrieval and data integrity.

8. Development Approach

The system can be developed using a modular approach, where each component (submission, tracking, management, feedback) is built and tested separately, then integrated into a complete system. Standard software development practices such as requirement analysis, design, implementation, testing and maintenance are followed.

Requirement Analysis

Functional Requirements

1. User Registration and Secure Login
1.1 Citizens can create accounts using name, phone number, email, and national ID.
1.2 The system must implement secure authentication password encryption.
1.3 Only authorized users can log in and submit complaints, ensuring personal data protection.
2. Submit Complaints with Category, Description, Attachments
2.1 Citizens can select relevant service categories (e.g., roads, electricity, water).

2.2 The complaint form must allow detailed descriptions of the issue.

3. Real-Time Complaint Tracking

3.1 The system generates a unique complaint ID after submission.

3.2 Citizens can track status updates such as “Pending,” “In Progress,” or “Resolved.”

4. Post-Resolution Feedback Submission

4.1 After resolution, citizens are invited to provide feedback.

4.2 Feedback may include satisfaction ratings and written reviews.

4.3 Citizens can request to reopen the complaint if unsatisfied.

5. View Assigned Complaints

5.1 Government officers can view complaints assigned to their department or role.

5.2 Complaints can be filtered by date, priority, category, or location.

5.3 Dashboards display workload and pending responsibilities.

6. Update Complaint Status

6.1 Officers can update the complaint status as work progresses.

6.2 Status changes are instantly reflected in the citizen’s tracking interface.

6.3 This ensures transparency and keeps citizens informed.

7. Prioritize Urgent Complaints

7.1 Officers can mark complaints as high priority based on severity or risk.

7.2 Urgent issues (e.g., water leakage, electrical hazards) are fast-tracked.

7.3 Priority tagging ensures critical problems receive immediate attention.

8. Add Resolution Details

8.1 Officers must submit a resolution report after resolving the issue.

8.2 The report includes actions taken, materials used, and completion date.

9. Manage Users

9.1 Admins can create, edit, suspend, or delete user accounts.

9.2 Role-based access control must be enforced for citizens and officers.

9.3 This ensures system security and proper user management.

10. Manage Departments

10.1 Admins can create and manage government departments.

10.2 Each complaint category is mapped to a responsible department.

10.3 This ensures automatic and accurate complaint routing.

11. Assign Complaints

11.1 Admins can manually assign or reassign complaints.

11.2 This is useful when auto-routing fails or workload balancing is needed.

11.3 Proper assignment helps speed up resolution time.

12. Generate Statistical Reports

12.1 The system generates reports on received, resolved, and pending complaints.

12.2 Department performance can be analyzed.

Non-functional Requirements

Security
- Secure authentication
- Data protection
- Role-based access control

Performance
- Instant submission processing
- Real-time updates
- Multi-user handling capability

Usability
- User-friendly interface
- Easy navigation
- Minimal training required

Availability
- 24/7 web access
- Desktop & mobile support

Reliability
- Accurate record maintenance
- Regular data backups

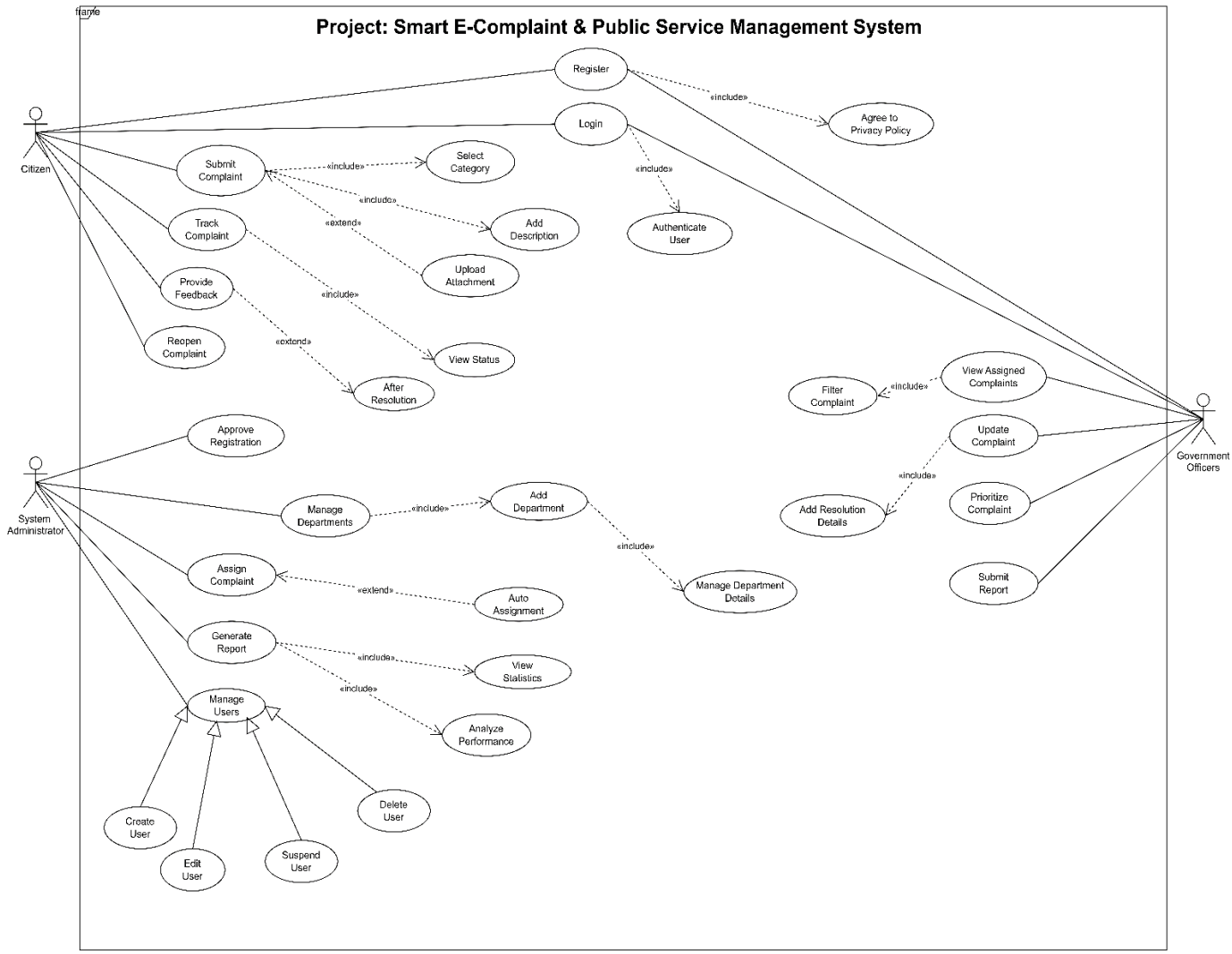
Maintainability
- Modular system design
- Easy upgrades and feature expansion

Political
- Follow government rules and policies
- Transparency in complaint handling to reduce corruption

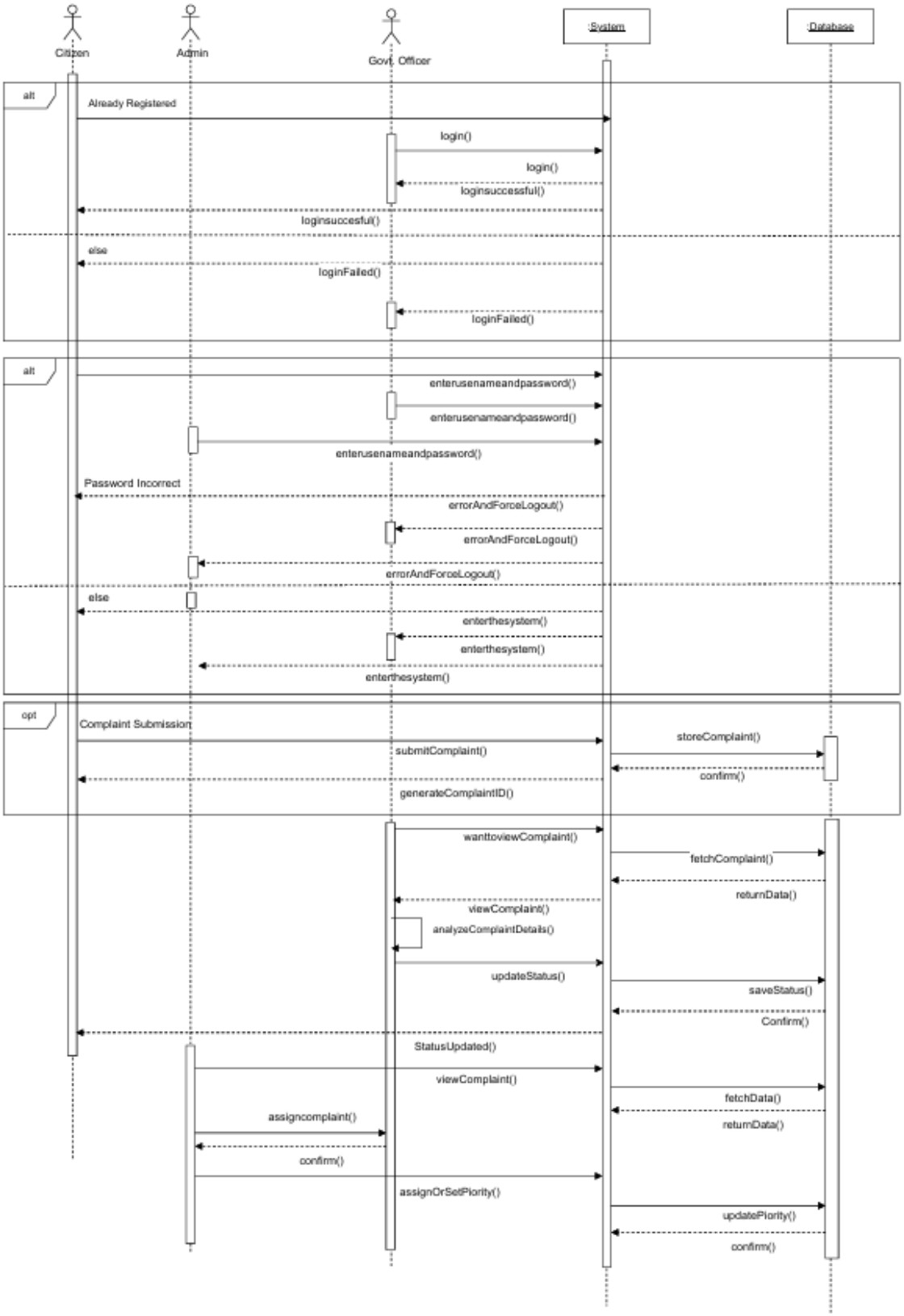
Cultural
- Simple design for all types of users
- Respectful and formal communication tone

Diagrams:

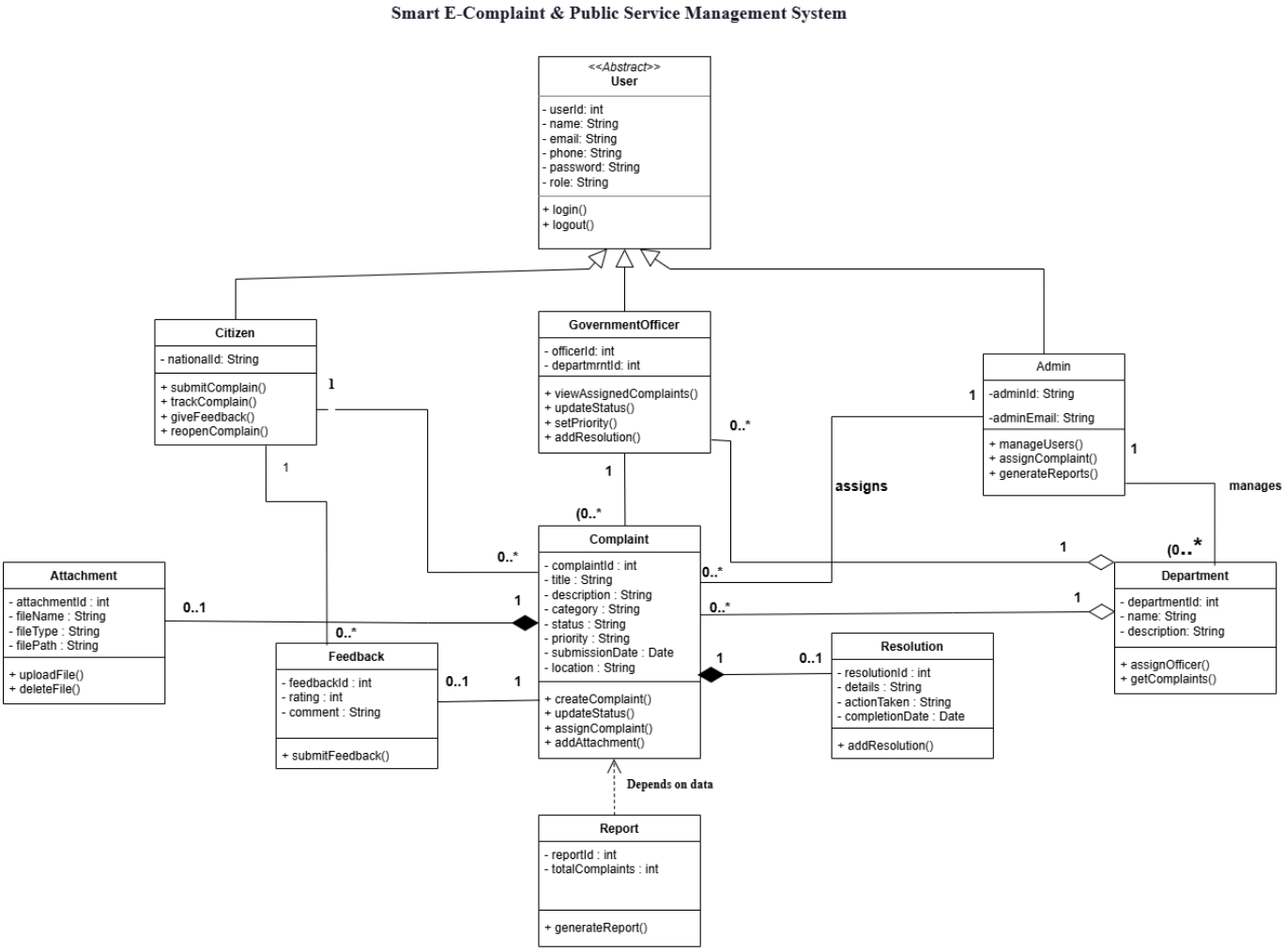
Use Case Diagram:



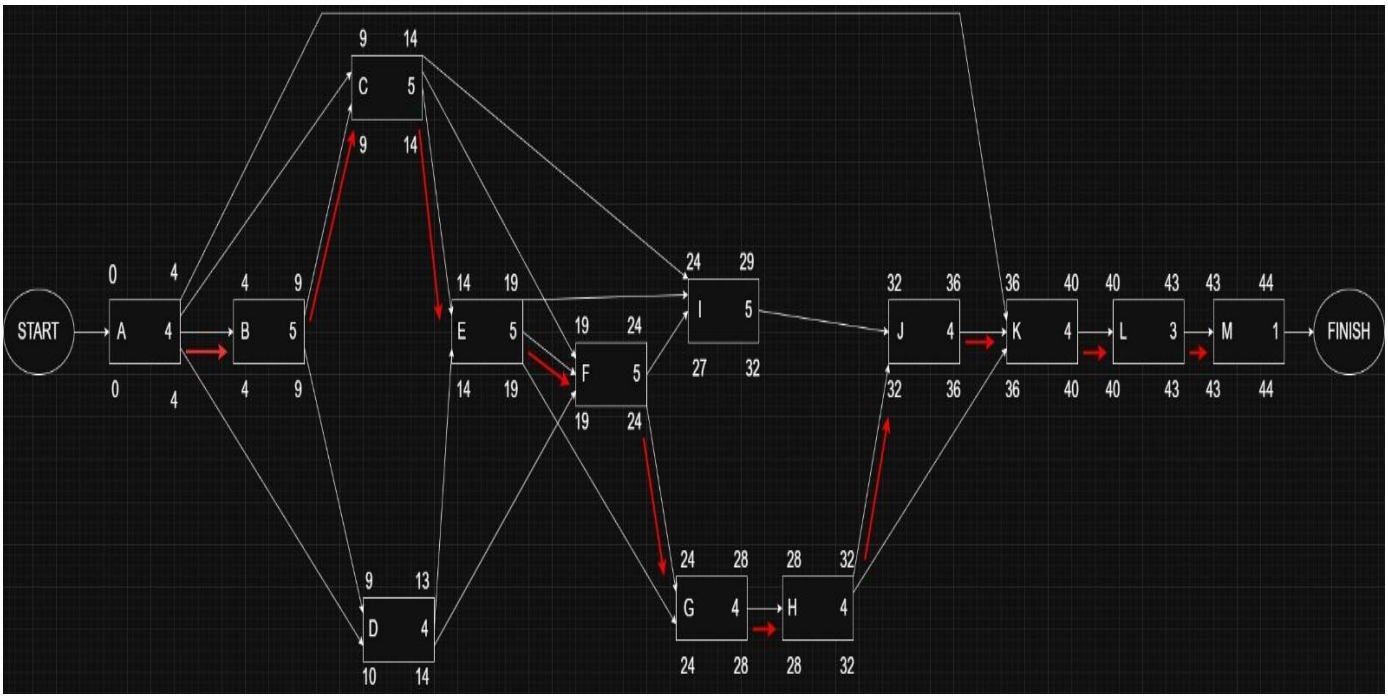
Sequence Diagram-



UML Class Diagram-



CPM Network Diagram:



Calculating Total Float (TF):

Formula: **TF = LS - ES or LF – EF**

Task	TF
A	0-0 = 0
B	4-4 = 0
C	9-9 = 0
D	10-9 = 1
E	14-14 = 0
F	19-19 = 0
G	24-24 = 0
H	28-28 = 0
I	27-24 = 3
J	32-32 = 0
K	36-36 = 0
L	40-40 = 0
M	43-43 = 0

Free Float (FF)

Formula: **FF = Min ES (next task) – EF (current task)**

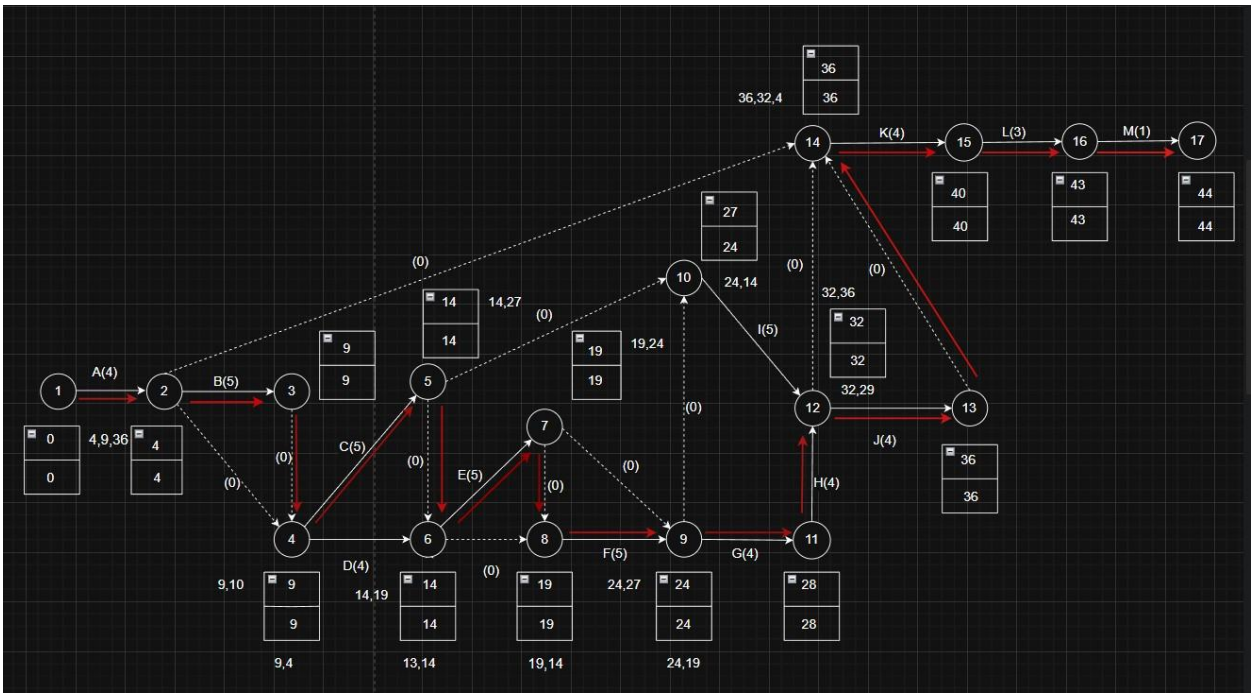
Task	TF
A	min(4)-4 = 0
B	min(9)-9 = 0
C	min(14)-14 = 0
D	min(14)-13 = 1
E	min(19)-19 = 0
F	min(24)-24 = 0
G	min(28)-28 = 0
H	min(32)-32 = 0
I	min(32)-29 = 3
J	min(36)-36 = 0
K	min(40)-40 = 0
L	min(43)-43 = 0
M	0

Critical Path:
A → B → C → E → F → G → H → J → K → L → M

Project length:

4+5+5+5+5+4+4+4+4+3+1 = 44 days

Pert Network Diagram:



Critical Path:
A → B → C → E → F → G → H → J → K → L → M

Project length:
4+5+5+5+5+4+4+4+4+3+1 = 44 days

Task	Dependency	Optimistic (to)	Most Likely (tm)	Pessimistic (tp)	Expected Time (te)	Variance (σ^2)
A	--	1	3	11	4	2.78
B	A	3	4	11	5	1.78
C	A,B	2	4	12	5	2.78
D	A,B	1	3	11	4	2.78
E	C,D	3	5	7	5	0.44
F	C,D,E	1	4	13	5	4
G	F,E	1	3	11	4	2.78
H	G	2	3	10	4	1.78
I	C,E,F	1	2	21	5	11.11
J	H,I	2	4	6	4	0.44
K	A,H,I,J	2	3	10	4	1.78
L	K	1	3	5	3	0.44
M	L	1	1	1	1	0.00

Critical Path:
A → B → C → E → F → G → H → J → K → L → M

Expected Project Length:
4 + 5 + 5 + 5 + 5 + 4+ 4 + 4 + 4 + 3 + 1 = 44 days

Project Length variance, σ^2 = 2.78+1.78+2.78+0.44+4+2.78+1.78+0.44+1.78+0.44+0.00=19

Project Length standard deviation, σ = 4.36

Now, $z = \frac{ts-te}{\sigma}$

Where, ts - the schedule time to complete the project

te - Normal expected project length

σ - Expected standard deviation of the project

Now, Let's check probability of completing the project in 50 days

The probability that the project will be completed in Ts = 50 days is given by $P(z \leq D)$

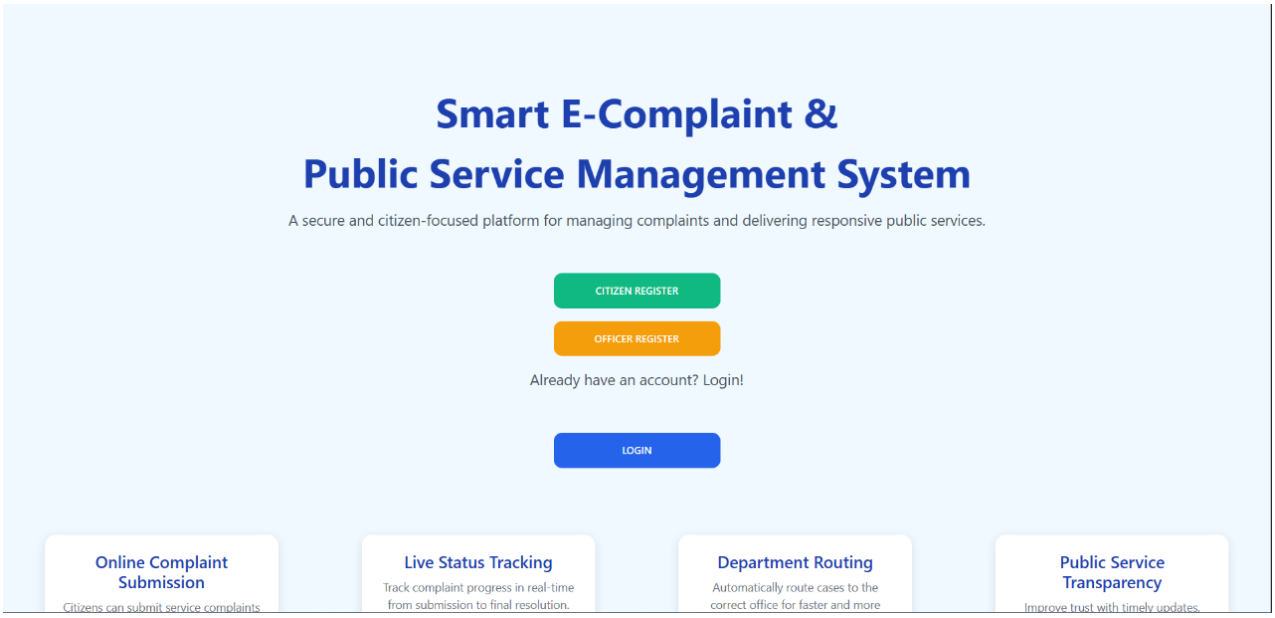
$$D = \frac{ts - te}{\sigma} = \frac{50 - 44}{4.36} = 1.38$$

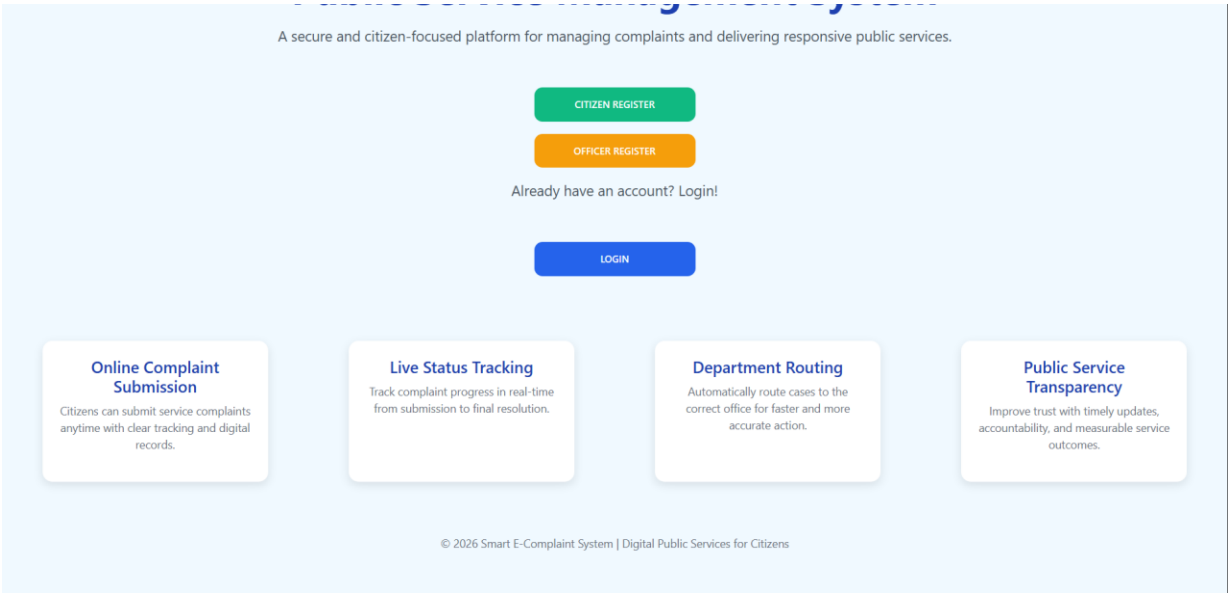
$$P(z \leq 1.38) = 0.91621 = 91.621\%$$

The probability that the project will be completed in Ts = 50 days is 91.621%

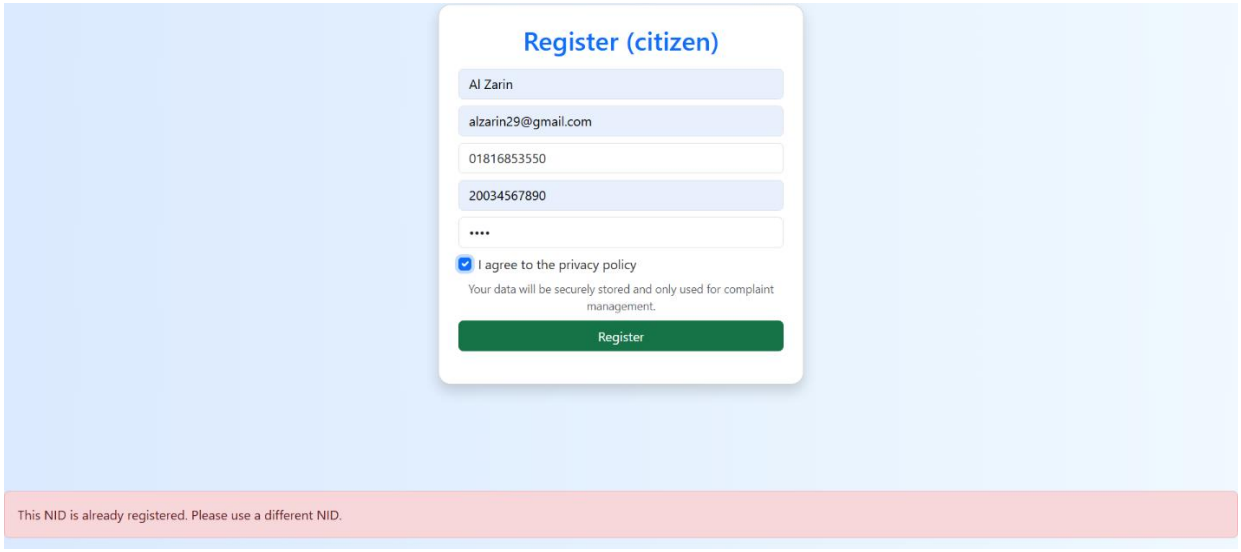
Screenshots of UI

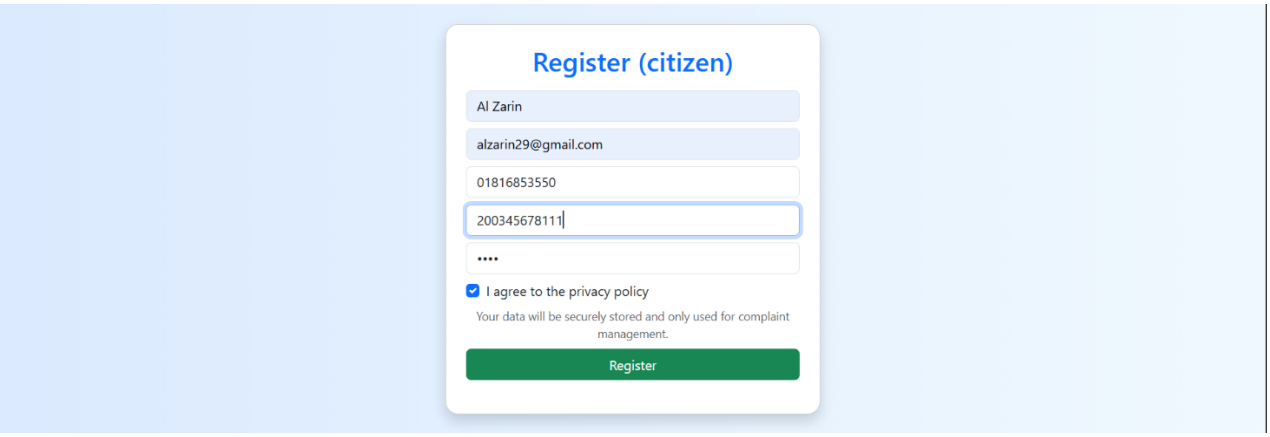
1. This is the homepage of our Smart E-Complaint & Public Service Management System, where citizens and officers can manage and track public service complaints.

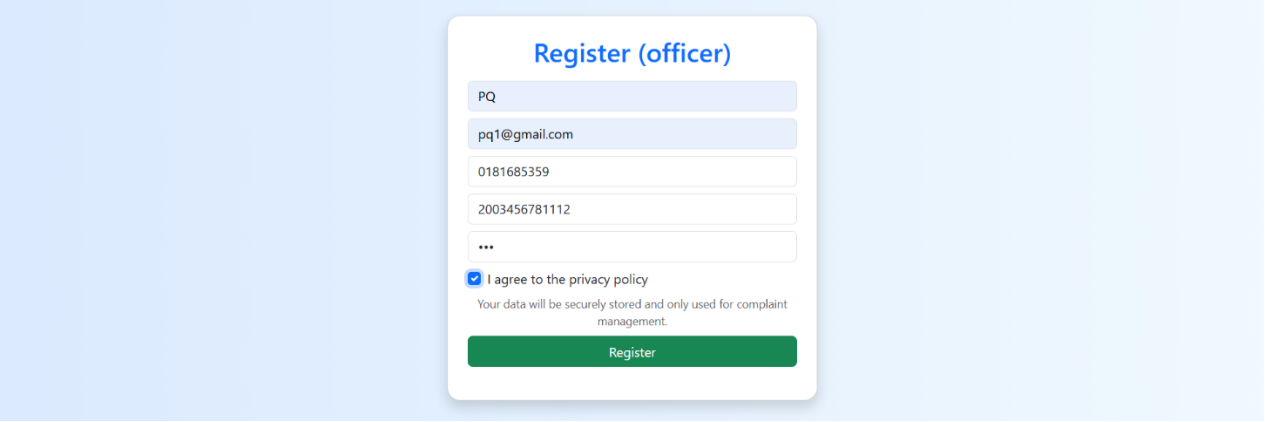




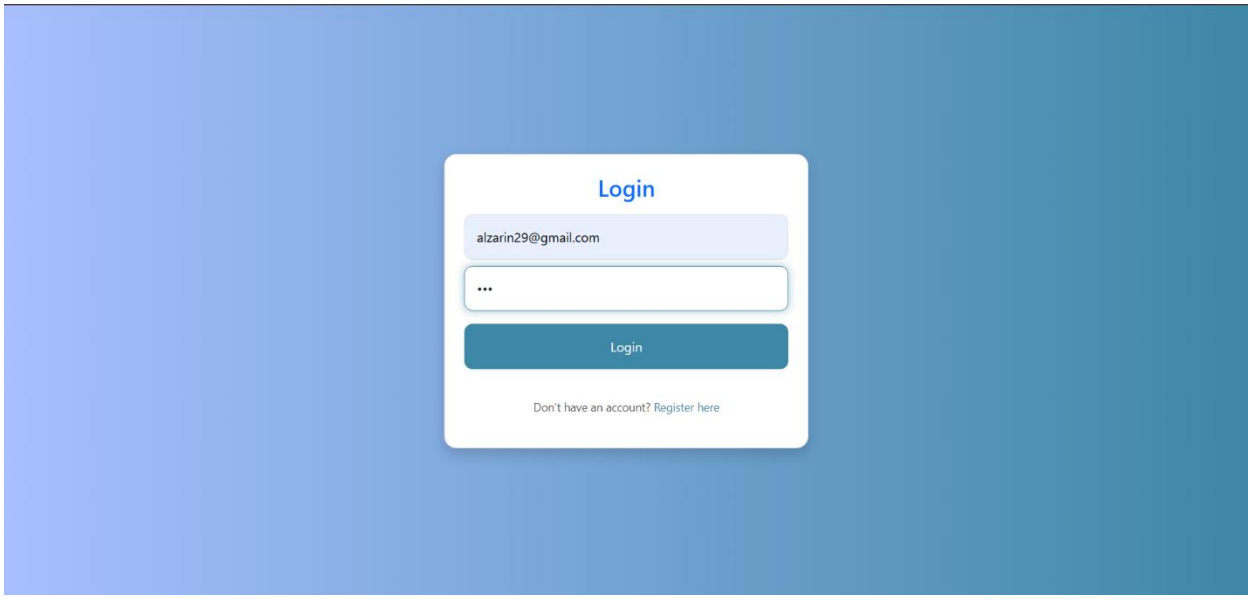
2. Here, citizens and officers can easily register by providing their personal details and agreeing to the privacy policy for secure handling of their data. If the entered NID is already registered, the system will prompt users to use a different one to proceed with the registration.



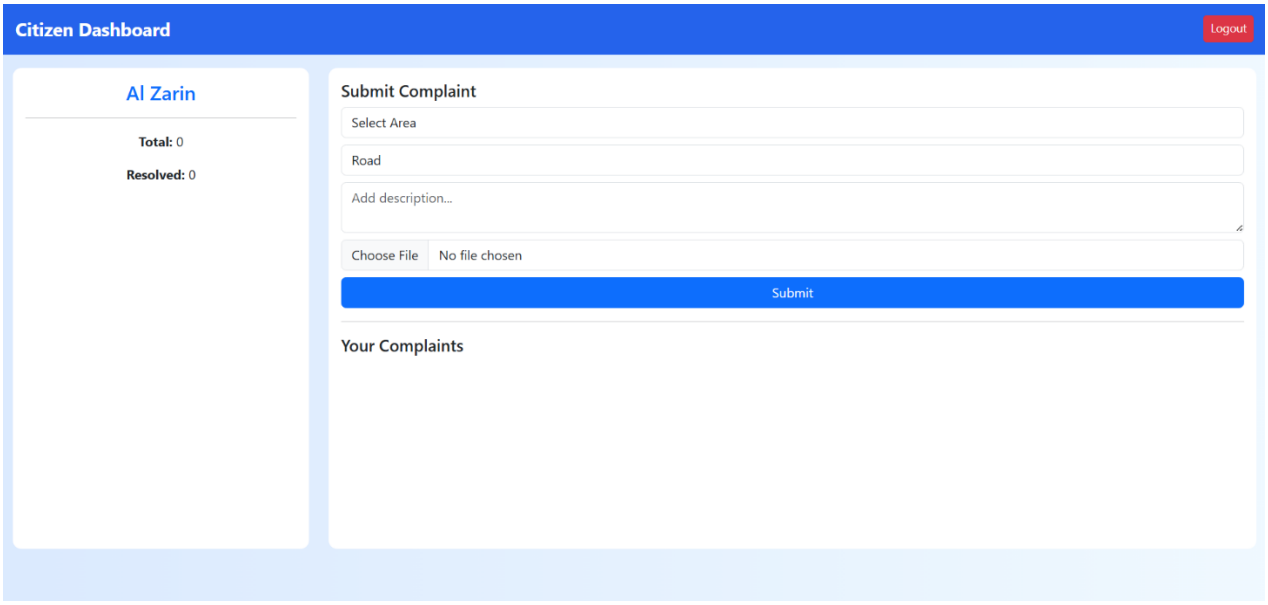




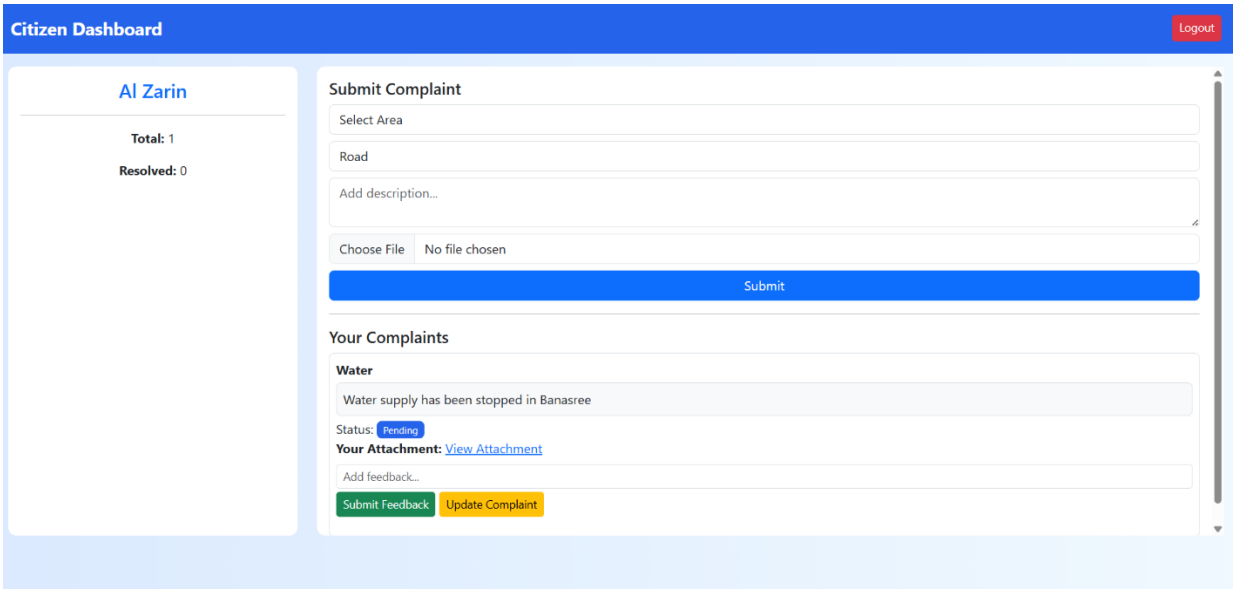
3. Existing users can log in using their email and password to access the platform and submit or track their complaints.



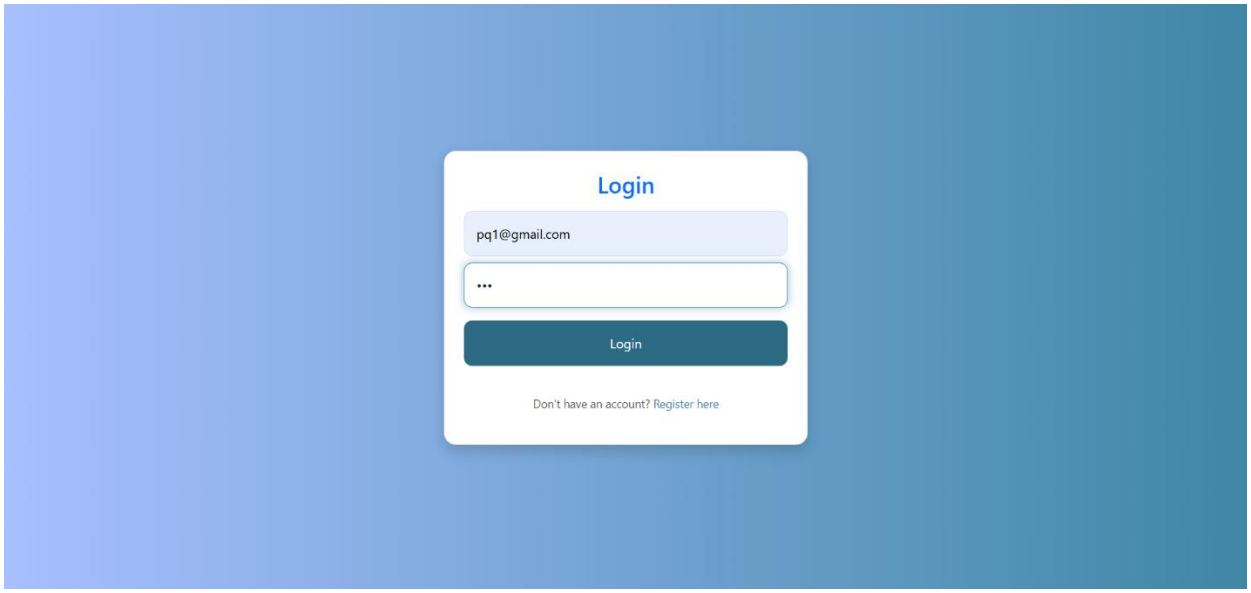
4. The citizen dashboard provides an overview of submitted complaints, allowing citizens to monitor the status of their cases.



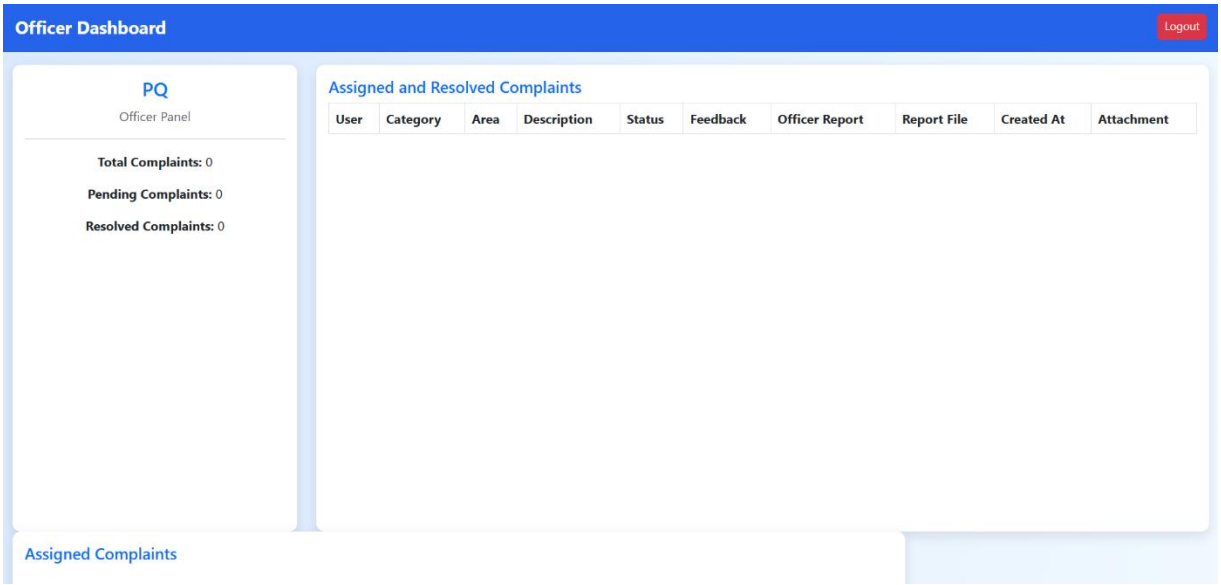
5. Citizens can submit new complaints, selecting the relevant category, area, and adding additional details or attachments.



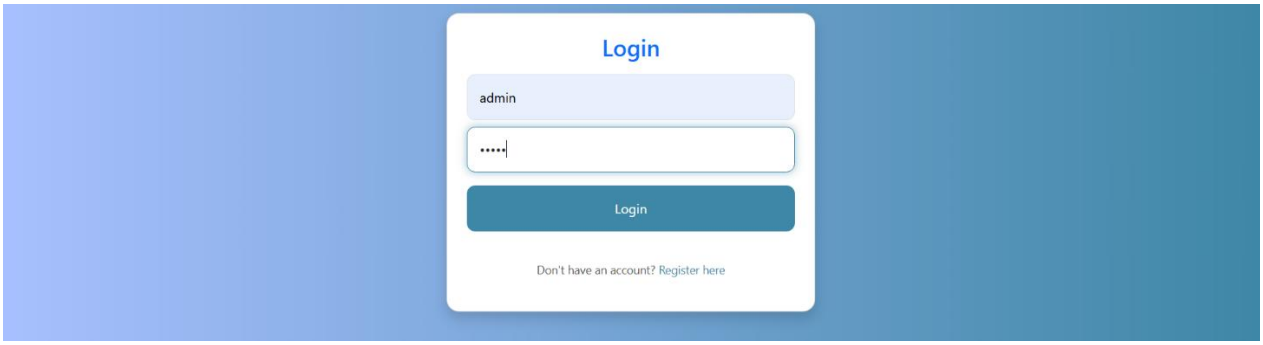
6. Officers can log in using their credentials to view assigned complaints, update their status, and resolve issues efficiently.

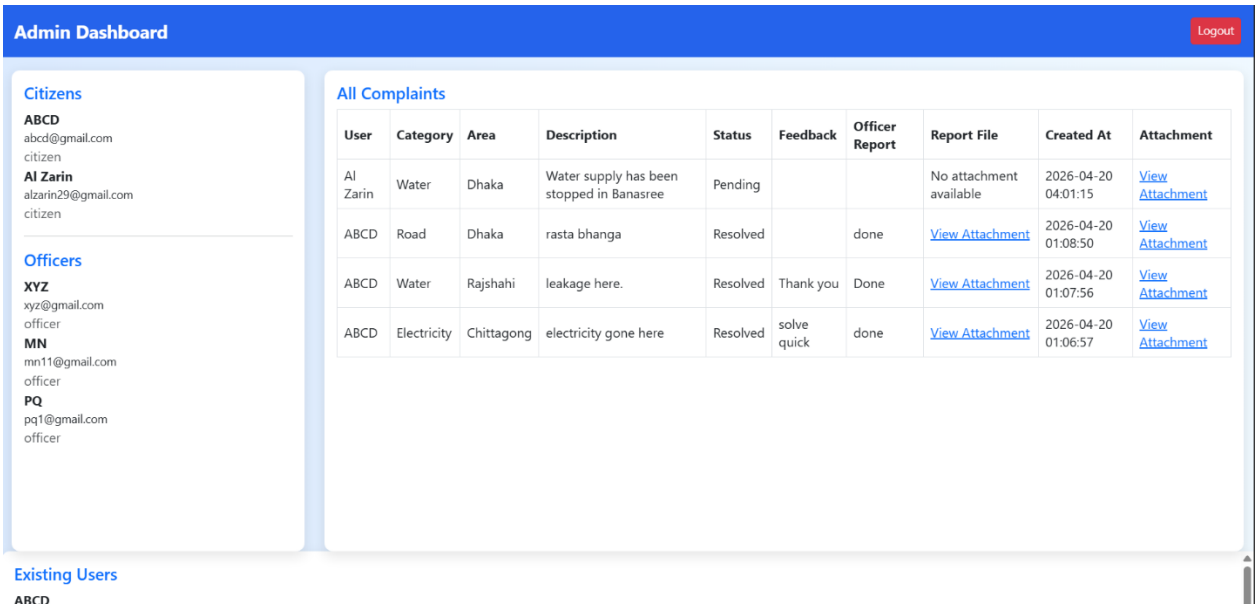


7. The officer dashboard shows the list of assigned complaints, allowing officers to track and resolve them in a timely manner.

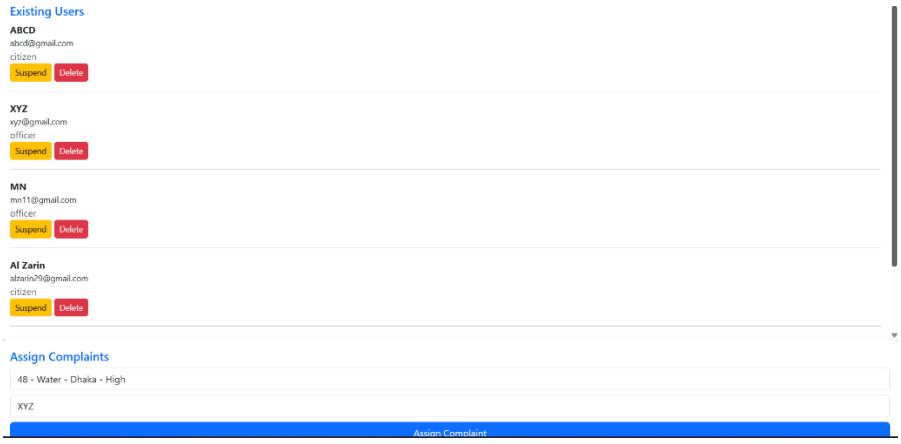


8. The admin dashboard provides a comprehensive view of all complaints, including their status and the actions taken by officers..

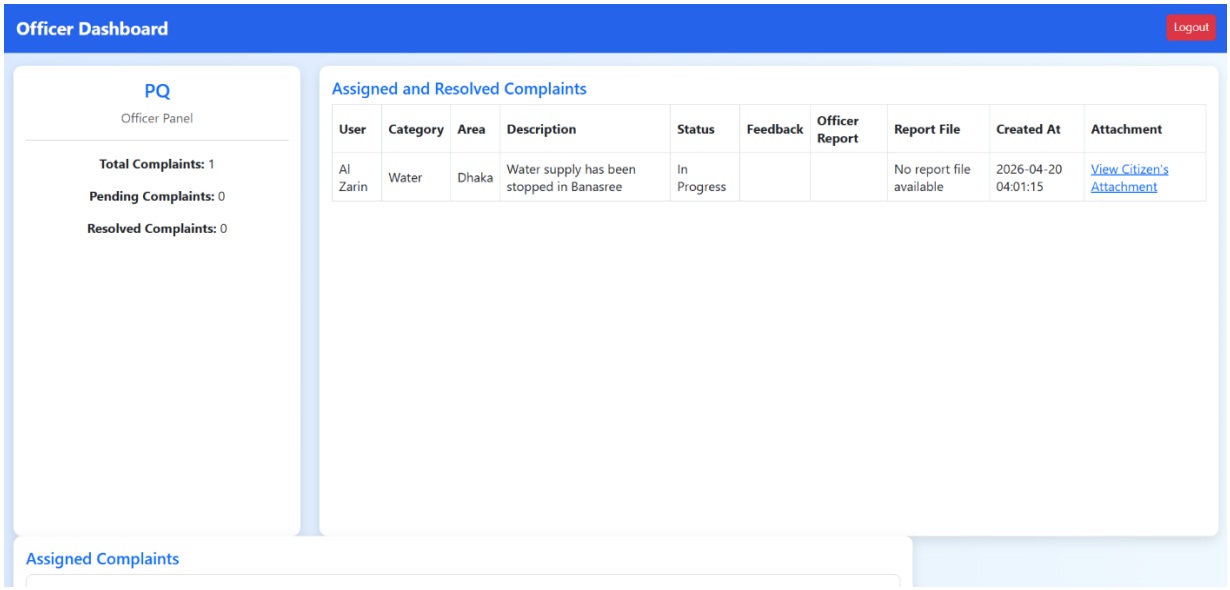




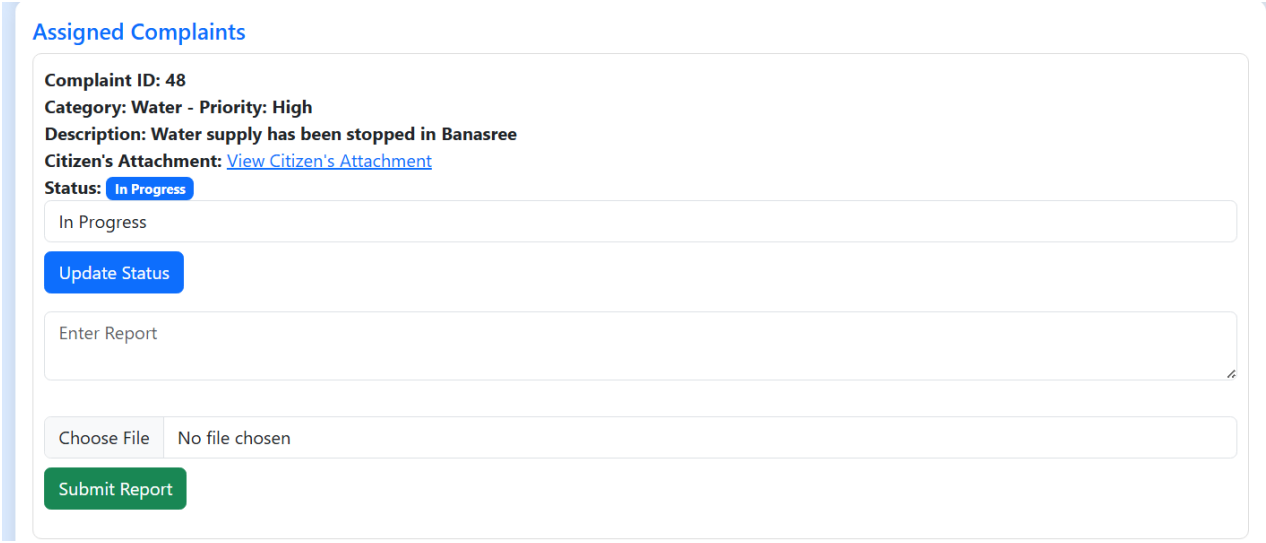
9. Admins can manage users by suspending or deleting accounts and assigning complaints to appropriate officers. He can assign complaints to specific officers based on the category and area, ensuring proper allocation of tasks.



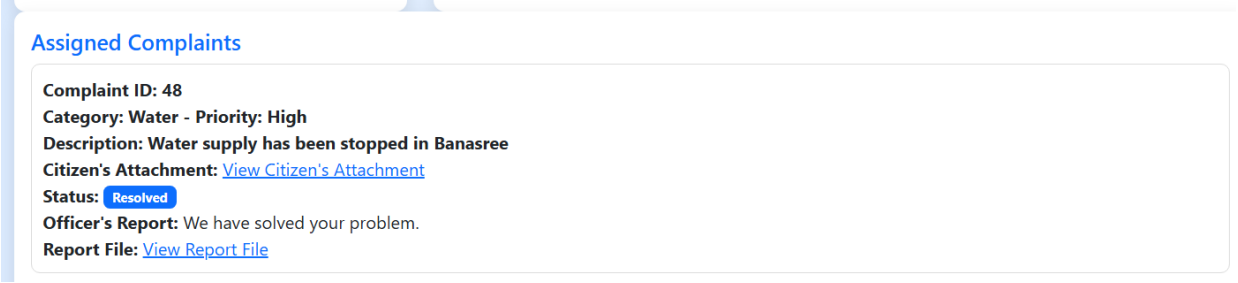
10. After assigning, the complaint is shown here in the assigned and resolved section of officer dashboard.



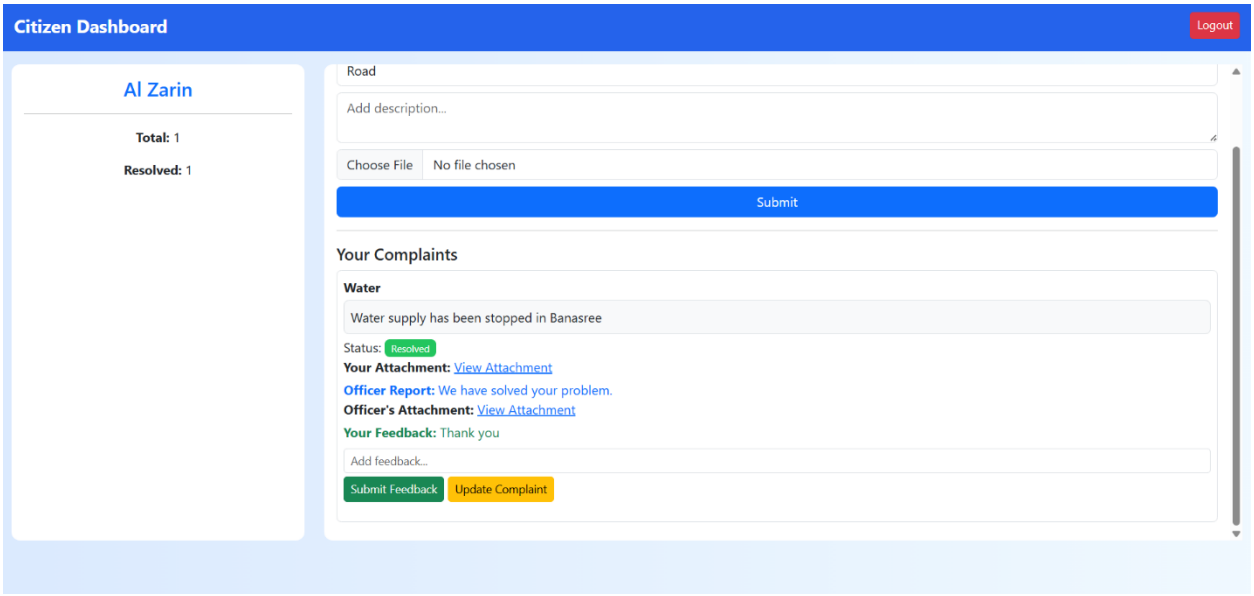
11. This section allows officers to view and manage complaints assigned to them, along with relevant details and citizen attachments



12. Officers can update the status of a complaint, upload reports, and submit resolutions after addressing the issues raised by citizens. Once a complaint is resolved, officers update the status to 'Resolved,' and citizens are notified with the final report and feedback options.



13. In citizen Dashboard, the complaint has been resolved now and citizen can see officer's report.



Conclusion

In conclusion, the Smart E-Complaint & Public Service Management System is an efficient, secure, and user-friendly platform that bridges the gap between citizens and public service officers. It empowers citizens to easily submit complaints, track their progress in real-time, and provide feedback on resolutions. Officers benefit from a streamlined process to manage, resolve, and report on complaints, ensuring transparency and accountability in public service. The system fosters trust and promotes responsiveness, contributing to improved public service delivery.